

Giovanni Battista Pernazza

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EXPERIENCE

FREELANCE

Software Developer

Remote
Nov 2024 - present

EdgeWorks srl : Architected custom AI-driven solutions to automate critical business workflows and enhance operational efficiency:

- **Skillmatch**: Engineered an AI-powered recruitment platform using React, TypeScript, and Supabase with a decoupled Python backend. Integrated LLMs to parse PDF CVs and generate automated candidate rankings (0-100% match scores), significantly reducing screening time for recruiters.
- **MORA (Mail Orchestration & Reply Assistant)**: Architected a Python-based LLM automation system using a modular DagPipeline to classify emails and generate drafts with RAG integration, streamlining management of 1000+ monthly emails.
- **TimeTracker**: Modernized internal project tracking by engineering a full-stack application using **React 19**, **TypeScript**, and **Supabase**. Designed a high-performance, fully responsive UI/UX with secure authentication and integrated **Recharts** for real-time analytics. Implemented automated Excel exports, streamlining financial reporting and resource management.
- **Archibit**: Architected an enterprise-grade Learning Management System for professional accreditation using **Next.js 16** (App Router), React 19, and Supabase. Engineered a secure, anti-cheat video playback engine powered by **Mux Video** for high-performance streaming, implementing visibility tracking and playback rate locking to ensure compliance for Professional Training Credits (CFP). Automated administrative workflows, including instant **PDF certification** generation and **granular progress tracking** for authorized training centers
- **Marsilea**: Reduced administrative overhead by 70% by architecting and deploying a full-stack Google Workspace web app (Apps Script, Sheets) that automated timesheet management and financial reporting for a team of 10.

Associazione Recup: Engineered an interactive dashboard with Recharts that automated KPI calculation (CO2, water saved), reducing manual reporting time for stakeholders by 95% (from hours to minutes).

SAPIENZA FOILING TEAM

Software Engineer

Rome, Italy
Sept 2024 - present

Tech Stack: C++, ESP32, GPS, IMU, SD Card, Python, Next.js, React, TypeScript, Supabase, Tailwind CSS, Recharts

- Architected embedded firmware for an ESP32-based telemetry system integrating GPS (10 Hz), IMU (100 Hz), and SD card logging, capturing **360,000+ synchronized sensor samples per hour** with millisecond-precision timestamps.
- Engineered an end-to-end telemetry pipeline transforming raw sensor streams into structured CSV datasets and powering a React/Next.js analytics dashboard for visualization of GPS tracks, speed, heading, acceleration, and inertial data across sailing sessions.
- Reduced post-session performance analysis time by **80%** by automating telemetry collection, processing, and visualization, accelerating engineering iteration and athlete performance evaluation.
- Developed and maintained the team's website and CMS, achieving a **95+ Lighthouse** performance score, increasing user engagement by **60%**, and enabling non-technical members to independently manage website content.

EDUCATION

UNIVERSITÀ LA SAPIENZA

Bachelor of Science in Computer Engineering

Rome, Italy
2021 - expected 2026

- **Relevant Courses**: Data Structures & Algorithms, Software Design (OOP), Computer Systems & Concurrency, Telecommunications

Phillips Exeter Academy

- Attended an international summer program focused on academic excellence, cultural exchange, and leadership development.

Exeter, NH, US
2017

SIDE PROJECTS

Self-Hosted Home Server & Private Cloud Infrastructure

Tech Stack: Linux, Docker, Docker Compose, Tailscale, Python, PostgreSQL, Pi-hole, Immich, Netdata

- Designed and deployed a self-hosted private cloud infrastructure on Linux, running containerized services with Docker Compose, secure remote access through Tailscale VPN, and automated monitoring workflows.
- Built and maintained a production-like environment with photo storage, DNS filtering, system monitoring, uptime tracking, and custom Telegram-based server health alerts, improving reliability and operational visibility.

FoxRun - Strava Performance Analyzer

Tech Stack: Python, FastAPI, React, TypeScript, Pandas, Tailwind CSS

- Delivered real-time athletic performance analysis by architecting a full-stack application that synchronizes with the Strava API via OAuth2, processing data with a Python/FastAPI backend.
- Developed an analytics dashboard with React, TypeScript, and Recharts to visualize performance history and trends.

SKILLS

Languages: Python, JavaScript, TypeScript, C++, C, Java, Ruby, SQL

Frameworks & Libraries: React, Next.js, Node.js, FastAPI, Flask, Ruby on Rails, Pandas

Databases: PostgreSQL, Supabase, NoSQL

Tools & Platforms: Git, Linux, Docker, AWS, Electron, ns-3, Wireshark, Jest

Languages: Italian (Native), English (C1 Advanced), French (A2 Basic)